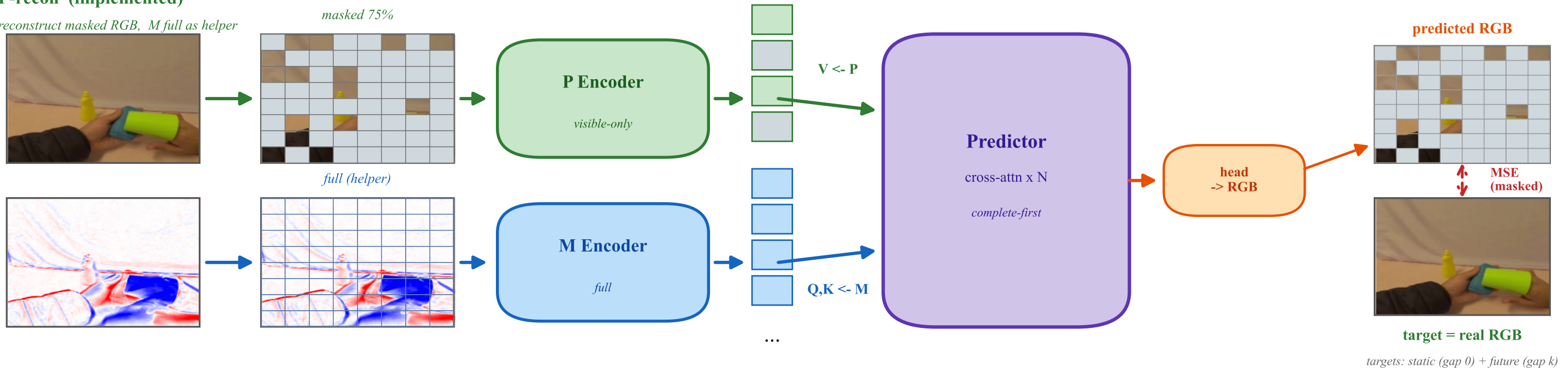


CoMP-MAE — Co-reconstructive Magno-Parvo MAE

Each stream reconstructs its own masked signal using the other as a full helper (V = target owns content, Q, K = helper) — code v16

P-recon (implemented)

reconstruct masked RGB, M full as helper

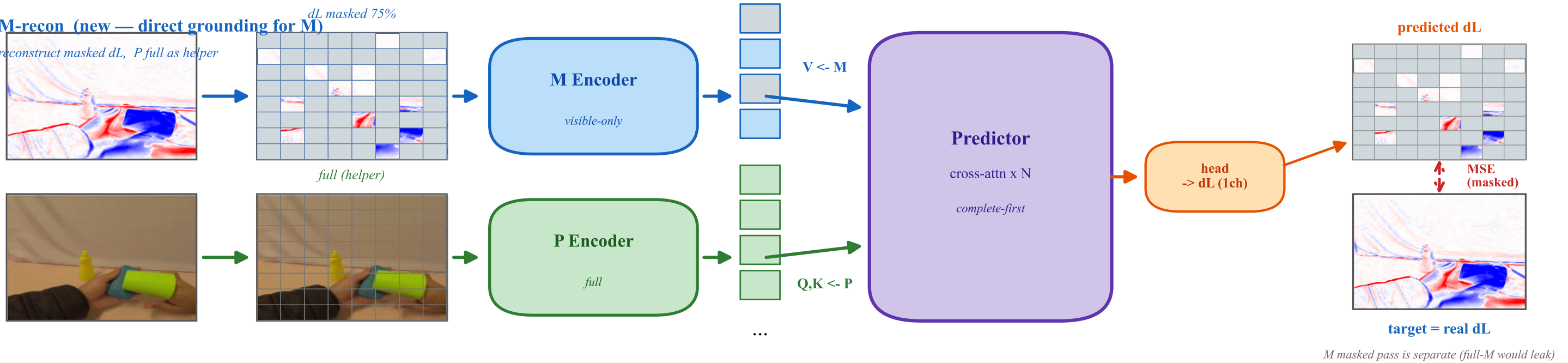


Cross-attention principle: V = reconstruction target (owns content -> the encoder must learn it) · Q, K = helper (low-bandwidth routing pattern only)

structure is mirrored, meaning is asymmetric: P -recon = motion correspondence | M -recon = form grouping (a natural consequence of P/M information asymmetry)

M-recon (new — direct grounding for M)

reconstruct masked dL, P full as helper



Symmetric Co-reconstruction

each stream: own signal masked / partner full -> cross-condition

$$L = L_P(\text{RGB}) + \lambda_M * L_M(\text{dL}) \quad (\text{both: gap } 0 + \text{gap } k)$$

What changes (vs MCP-MAE / MotionMAE)

MCP-MAE : P -recon only -> M trained as router only (no-op risk)

CoMP-MAE: + M -recon -> M grounded by pixels + mutual coupling